The Potential Role of Anomalous Geothermal Flux for Enhanced Basal Melting and Suppressed Ice Velocity at Haynes Glacier, West Antarctica

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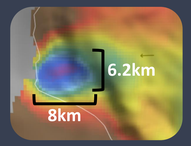
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Introduction

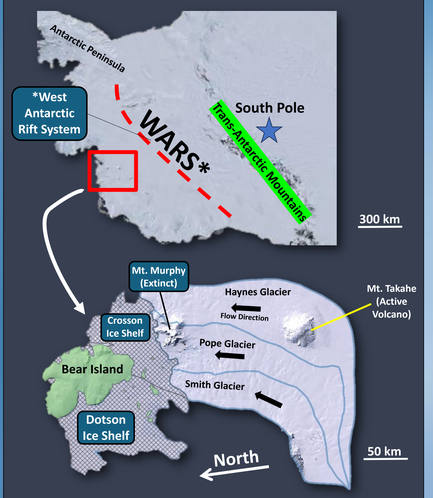
- Novel Velocity Anomaly aka "Sticky Spot" at the Grounding Line of Haynes Glacier.
- High Intensity Dipping in Radar Layers Upstream Indicative of Enhanced Basal Melting.
- These Novel Phenomena are located within a region known for its High, Spatially Variable Geothermal Flux and Subaerial Volcanism.

Sticky Spot

- Ice moving >300 m/yr slows abruptly at the Grounding Line
- 8 km x 6.2 km, oval shaped



Where Are We?



1

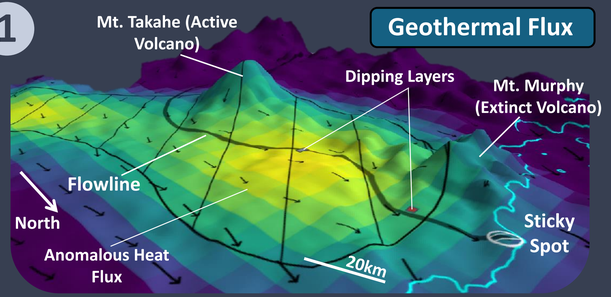


Figure 1: Geothermal Flux interpolated from Curie Depth. The Black Hash Lines denote a region of Anomalous Geothermal Flux under Haynes Glacier. Black Arrows Indicate Ice flow Direction. Data from Dziadek et al, 2021

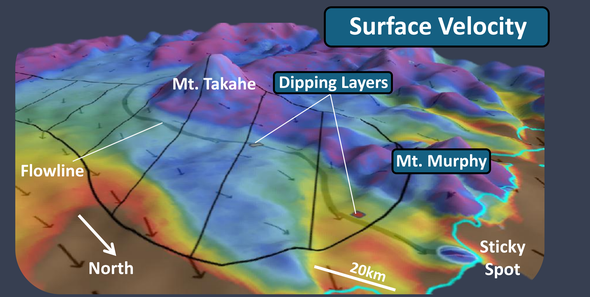
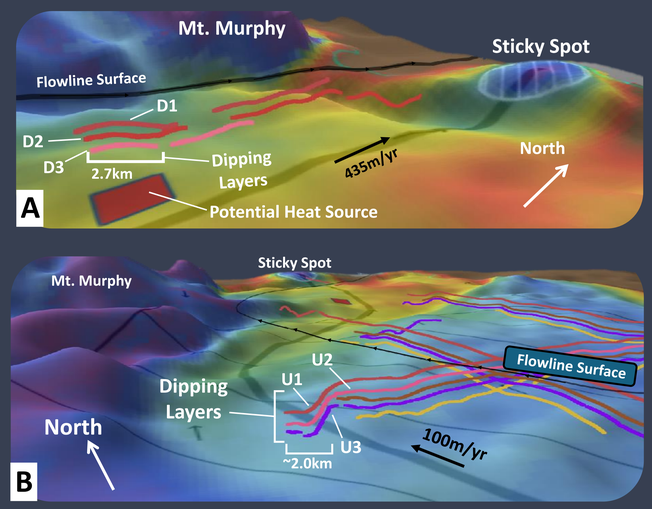
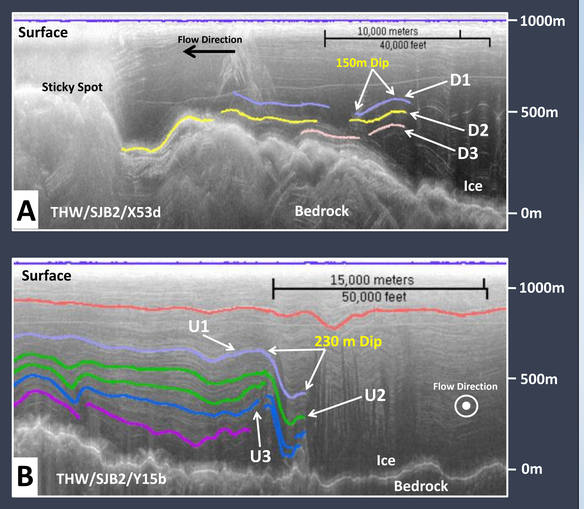
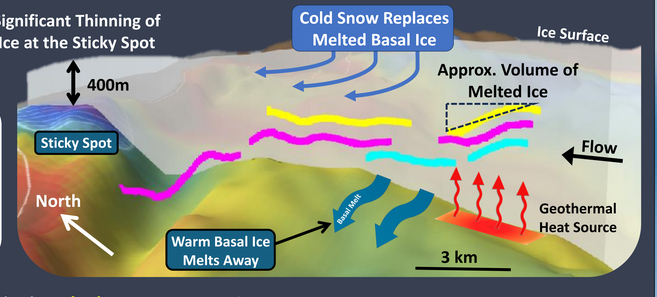


Figure 2: NASA's MEaSUREs Ice Surface Velocity Data. Black Hash Lines indicate the Zone of High Geothermal Flux from Figure 1. Arrows Indicate the Direction of Ice Flow.





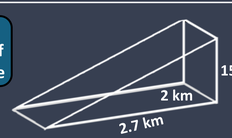
2



*Horizontal Velocity >> Vertical Velocity (~400 m/yr vs. 1-2 m/yr)

- Warm Basal ice is melted away
- Cold Accumulation replaces lost ice
- After thinning, the cooled ice stream no longer has enough thickness to induce pressure melting = stuck to bed

Approx. Volume of Melted Ice



Amount of Geothermal Flux Needed?

$$Q = V\rho_{ice}H_{fus} \qquad V = 0.405\ km^3$$

$$G = \frac{Q}{Area(time)} \qquad G \cong 140\ W/m^2$$

*Geothermal Flux of Active Volcanic Sites Can Reach Up To 1300 W/m²

Expanding our Area and Time Horizon...

$$G \cong 1.2\ W/m^2$$

*Lowball Estimate

3 What Does This Mean For West Antarctica and Beyond?

1. Diverse Evidence for Active Subglacial Volcanism in West Antarctica continues to grow.
2. Active Volcanism could play a larger role In Ice Sheet Stability and Evolution than we realize.
3. Uncertainties Are Still High - Is Ice Heated Intensely at a Point Source Underneath the Layers? Or is Broader Regional Heat a More Likely Cause?

Acknowledgements

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References